Case 02 · Ideation document: Rate-Shock Early Warning

Following chapter 1 (Initiation) of the Magshimim project-book template. For each project: overview, goal, target audience, existing solutions, technological core, technologies, challenges and sources, hardware requirements, and conclusion.

Overview

A system that listens in real time to public market data (yields, cross-country spreads, the term premium, surprises in economic releases), using a streaming data pipeline and anomaly detection, to warn of dangerous combinations before they become headlines.

In practice: the system continuously computes a set of indicators for each country, such as France's spread over Germany, the gap between the interest rate and growth, and the speed of change in the two-year yield. When a single indicator leaves its normal range, or when two move together in a dangerous direction (a widening spread while r overtakes g), an alert is sent with an explanation: what moved, by how much, and why it is worrying.

Project goal

We chose this project because the story has moments when the market "got" something all at once: on September 23rd, PMI day, the yield jumped 0.15 points, the biggest move of the year. The signs were there earlier, scattered across dozens of sources.

The goal: gather the signs in one place and detect dangerous combinations automatically, instead of relying on someone to connect them.

Target audience

- **Primary audience:** risk managers at small and mid-sized institutions (pension funds, insurers) without a large macro research team. Required knowledge: an understanding of the bond market.
- **Secondary audience:** economics journalists and sophisticated private investors.
- **Illustrative scenario:** a risk manager at a mid-sized pension fund gets a morning alert: the Italian spread has widened three days running while the European two-year yield jumps. She checks the fund's exposure to Italian bonds before the market opens.

Existing solutions

- **OFR Financial Stress Index (Office of Financial Research, US Treasury):** a daily, public and free index of financial stress that combines dozens of market variables (credit spreads, volatility, yield curves) and breaks stress down by source and region.
- **ECB CISS (Composite Indicator of Systemic Stress):** the ECB's systemic-stress indicator, public and regularly updated, built specifically to detect stress that appears in several markets at once.

- **St. Louis Fed Financial Stress Index (STLFSI):** a weekly, public stress index, available through FRED.
- **Bloomberg and Refinitiv:** every number in real time, with user-configured alerting, for subscribers.
- **IMF Early Warning Exercise:** a twice-yearly IMF exercise to identify systemic vulnerabilities.

What sets us apart: this is where the difficulty shows. The OFR, ECB and St. Louis Fed stress indices do exactly what we described (combining many signals into one number that warns of stress), and they are public, free, research-based and updated daily. What is left for us to add is mainly an alerting interface. That is not novelty; it is a display layer.

Technological core

The core is a real-time anomaly-detection engine: a pipeline that receives market events, computes moving windows for each indicator, and applies algorithms such as CUSUM (cumulative change detection) and Isolation Forest (multidimensional anomaly detection) to decide when to send an alert.

Main technologies

- **Redpanda (Kafka-compatible):** an event queue for market data.
- **Bytewax:** stream processing in Python, to compute moving windows.
- **scikit-learn:** for Isolation Forest.
- **FRED API and ECB Data Portal:** public data sources.
- **Grafana:** dashboards and alerting.

Open question: Apache Flink versus Bytewax. Flink is powerful and proven at scale; Bytewax is simpler and in Python. At the volume of public data, Flink is probably overkill.

Technological challenges and sources

- **Data frequency:** free public data are mostly daily, not real-time. "Real time" over daily data is mostly a name.
- **False alarms:** a system that alerts too often starts being ignored. It needs calibration on history.
- **Overlap with existing indices:** every indicator we build will be compared automatically with the OFR FSI and CISS. If it matches them, it is redundant. If it differs, we must explain why ours is the right one.
- **What happens after the alert:** the system detects, but does nothing itself. Whoever receives the alert must know what to do with it, and that lies outside the project's boundaries.

Sources: financialresearch.gov (OFR FSI methodology), the ECB's CISS documentation, fred.stlouisfed.org.

Hardware requirements

None. A small cloud server is enough for the volume of public data.

Conclusion

⛔ Eliminated. Not because of technical difficulty: the architecture is rich and fully buildable. It falls on the most basic ideation question, why the project needs to exist. The OFR, ECB and St. Louis Fed financial-stress indices already do exactly this, publicly, for free and on years of research. What is left to build is an alerting interface over daily data, and that does not justify a project. And even at best it only warns, with nothing to do about the warning.